SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED

LOCATION: TRAIN 3	**AREA: D**
PERMIT TO WORK No.: —	**TYPE OF SCAFFOLD:** A Frame
SCAFFOLD DRAWING No: PMS-WA-ARCO-033	
RISK CLASSIFICATION: ○ High ✓ Medium ○ Low	

REFERENCE STANDARDS

- BS EN 1991-1-4:2005 — Wind Actions
- BS EN 1993-1-1:2005 — Design of Steel Structures (Eurocode 3)
- BS EN 12811-1:2003 — Temporary Works Equipment — Scaffolds

SCAFFOLD TO BE INSPECTED & TAGGED BEFORE USE

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of an **A Frame** scaffold required for *valve lifting during the turnaround maintenance activity in train 3* with equipment tag **314LICA-006** . The scaffold system consists of a **1.8m × 1.8m × 4.0m** A-Frame scaffold with vertical bracing for sway resistance.

SAFE WORKING LOAD (SWL): 3.691 kN (376 kg)

TOTAL HORIZONTAL LOAD SUMMARY

Direction	Wind (kN)	Impact (kN)	TOTAL (kN)
X	2.02	1.107	**3.127**
Z	1.42	1.107	**2.527**

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CALCULATION SHEETS

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GENERAL NOTE

BS EN 12811-1, 4.2.1.2

Scaffold materials are in accordance with BS EN 12811 with minimum design strength of 210 N/mm². This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 – Design of Steel Structures).

Material Properties

48.3 mm outside diameter × 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6.**

Modelling Philosophy

BS EN 12811-1, 4.2.1.2

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint (KFX/KFZ = 11.92 kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot. See Appendix A for model views.

LOAD CASES

Load diagrams are shown in Appendix B.

Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Chain Hoist Load

BS EN 1991, 6.3.4.1

The equipment static weight is **301 kg ≈ 2.953 kN**. A **25% dynamic allowance** was applied for chain hoist operation.

Static weight = 301 kg × 9.81 / 1000 = **2.953 kN**
Chain Hoist Design Load (CL) = 2.953 × (1 + 0.25) = **3.691 kN**

Impact Load

BS EN 1991, 6.3.4.1

30% of the chain hoist design load was assumed as a horizontal impact load, applied at the hoist attachment point independently in X and Z directions.

Impact Load (ILX) = 3.691 × 0.30 = **1.107 kN** (X-axis)
Impact Load (ILZ) = 3.691 × 0.30 = **1.107 kN** (Z-axis)

Wind Load

EN 1991-1-4:2005

Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.05678 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

Total Wind Load — X Axis (WLX) = **2.02 kN**
Total Wind Load — Z Axis (WLZ) = **1.42 kN**

WIND LOAD CALCULATION — EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V_3 =	**36.0**	m/s
	Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity		
Dredger P. — How to Approximate	Convert 3-second gust to 10-minute mean wind velocity		
	V_3 / V_{3600} =	1.52	
	V_{600} / V_{3600} =	1.06	
	$V_{600} = (V_{600}/V_{3600}) \times (V_{3600}/V_3) \times V_3$ =	**25.11**	m/s
	Basic wind speed $V_b = V_{600}$ =	**25.11**	m/s
EN 1991-1-4:2005	Roughness Length (z_0) — Terrain Category 0	0.003	m
	Terrain category II ($z_{0,ii}$)	0.05	
EN 1991-1-4, Table 4.1	z_{min}	1.0	m
	z — height of structure under consideration Source: STAAD command geometry height	**4.0**	m
EN 1991-1-4, Sec 4.3	$C_{r0}(z)$ — Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	$C_o(z)$ — Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z_0/z_{0,ii}]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z \rightarrow$ Case 1 applies		
	Wind Turbulence		
	$C_r(z) = K_r \times \ln(z/z_0)$	1.1227	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	28.187	m/s
EN 1991-1-4, Sec 4.4	K_1	1.0	
	Turbulence intensity $I_v(z) = k_1 / [C_o(z) \times \ln(z/z_0)]$	0.139	
	Peak Velocity Pressure		
	Density of air ρ	1.25	kg/m³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7 \cdot I_v(z)] \times 0.5 \times \rho \times V_m(z)^2$	**979.6342**	N/m²
Wind Load = $C_f \times q_p \times A_f$			
C_f =		1.2	
q_p =		0.979634	kN/m²
A_f =	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		**0.05678**	**kN/m**

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use γ = **1.5**; SLS (characteristic) combinations use γ = **1.0**.

DL = Dead Load | **CL** = Chain Hoist | **ILX/ILZ** = Impact X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	**1.5DL + 1.5CL**	LC1 × 1.5 + LC2 × 1.5
8	**1.5DL + 1.5CL + 1.5ILX**	LC1 × 1.5 + LC2 × 1.5 + LC3 × 1.5
9	**1.5DL + 1.5CL + 1.5ILZ**	LC1 × 1.5 + LC2 × 1.5 + LC4 × 1.5
10	**1.5DL + 1.5WLX**	LC1 × 1.5 + LC5 × 1.5
11	**1.5DL + 1.5WLZ**	LC1 × 1.5 + LC6 × 1.5

LC No.	Combination	Component Loadings & Factors
12	**1.5DL + 1.5CL + 1.5WLX**	LC1 × 1.5 + LC2 × 1.5 + LC5 × 1.5
13	**1.5DL + 1.5CL + 1.5WLZ**	LC1 × 1.5 + LC2 × 1.5 + LC6 × 1.5
14	**DL + CL**	LC1 × 1.0 + LC2 × 1.0
15	**DL + CL + ILX**	LC1 × 1.0 + LC2 × 1.0 + LC3 × 1.0
16	**DL + CL + ILZ**	LC1 × 1.0 + LC2 × 1.0 + LC4 × 1.0
17	**DL + WLX**	LC1 × 1.0 + LC5 × 1.0
18	**DL + WLZ**	LC1 × 1.0 + LC6 × 1.0
19	**DL + CL + WLX**	LC1 × 1.0 + LC2 × 1.0 + LC5 × 1.0
20	**DL + CL + WLZ**	LC1 × 1.0 + LC2 × 1.0 + LC6 × 1.0

STATIC CHECK PARAMETERS — UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.39 at Member No. **121**, Load Combination **10**, Clause **EC-6.3.3-662**. **PASS** — All members adequate; UC ratio below unity.

UC ratio diagram: see Appendix C.

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
121	PIP483.6	PASS	EC-6.3.3-662	**0.390**	10	1.00	C
28	PIP483.6	PASS	EC-6.2.9.1	**0.321**	12	0.67	T
24	PIP483.6	PASS	EC-6.2.9.1	**0.307**	10	0.16	T
38	PIP483.6	PASS	EC-6.3.3-662	**0.279**	12	0.96	C
96	PIP483.6	PASS	EC-6.2.9.1	**0.264**	8	4.98	C
104	PIP483.6	PASS	EC-6.2.9.1	**0.247**	8	5.12	C
90	PIP483.6	PASS	EC-6.3.3-662	**0.241**	9	1.92	C
105	PIP483.6	PASS	EC-6.2.9.1	**0.235**	13	0.25	T
102	PIP483.6	PASS	EC-6.2.9.1	**0.234**	9	0.15	T
89	PIP483.6	PASS	EC-6.3.3-662	**0.232**	9	1.94	C
86	PIP483.6	PASS	EC-6.3.3-662	**0.231**	8	1.77	C
45	PIP483.6	PASS	EC-6.2.9.1	**0.225**	8	0.00	C
112	PIP483.6	PASS	EC-6.2.9.1	**0.225**	8	0.00	C
80	PIP483.6	PASS	EC-6.2.9.1	**0.224**	9	0.77	C
95	PIP483.6	PASS	EC-6.2.9.1	**0.224**	8	1.06	C
99	PIP483.6	PASS	EC-6.2.9.1	**0.222**	9	0.16	T
69	PIP483.6	PASS	EC-6.2.9.1	**0.218**	9	2.11	T
59	PIP483.6	PASS	EC-6.2.9.1	**0.217**	12	0.02	T
65	PIP483.6	PASS	EC-6.2.9.1	**0.213**	8	0.36	T
82	PIP483.6	PASS	EC-6.2.9.1	**0.213**	8	0.61	C
75	PIP483.6	PASS	EC-6.2.9.1	**0.212**	8	4.95	T
111	PIP483.6	PASS	EC-6.2.9.1	**0.209**	12	0.04	T
68	PIP483.6	PASS	EC-6.2.9.1	**0.207**	9	2.25	T
15	PIP483.6	PASS	EC-6.3.3-662	**0.206**	12	7.82	C
108	PIP483.6	PASS	EC-6.2.9.1	**0.204**	13	0.23	T
103	PIP483.6	PASS	EC-6.2.9.1	**0.203**	8	5.10	T
85	PIP483.6	PASS	EC-6.3.3-662	**0.188**	9	1.47	C
78	PIP483.6	PASS	EC-6.2.9.1	**0.187**	12	0.62	C
74	PIP483.6	PASS	EC-6.2.9.1	**0.184**	8	1.04	T
49	PIP483.6	PASS	EC-6.2.9.1	**0.178**	12	0.04	T
57	PIP483.6	PASS	EC-6.3.3-662	**0.172**	10	4.10	C

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
64	PIP483.6	PASS	EC-6.2.9.1	**0.170**	8	0.98	T
13	PIP483.6	PASS	EC-6.2.9.1	**0.168**	10	4.35	T
40	PIP483.6	PASS	EC-6.3.3-662	**0.167**	10	0.47	C
76	PIP483.6	PASS	EC-6.2.9.1	**0.159**	13	0.64	C
124	PIP483.6	PASS	EC-6.3.3-661	**0.154**	9	1.89	C
101	PIP483.6	PASS	EC-6.2.9.1	**0.151**	8	2.69	C
36	PIP483.6	PASS	EC-6.3.3-661	**0.147**	12	5.81	C
110	PIP483.6	PASS	EC-6.2.9.1	**0.145**	8	3.03	C
17	PIP483.6	PASS	EC-6.2.9.1	**0.138**	12	1.12	C
61	PIP483.6	PASS	EC-6.3.3-662	**0.136**	8	1.09	C
97	PIP483.6	PASS	EC-6.2.9.1	**0.136**	9	0.77	T
98	PIP483.6	PASS	EC-6.2.9.1	**0.136**	9	1.62	C
100	PIP483.6	PASS	EC-6.2.9.1	**0.135**	9	2.99	T
125	PIP483.6	PASS	EC-6.3.3-661	**0.132**	10	1.56	C
19	PIP483.6	PASS	EC-6.3.3-662	**0.130**	12	3.38	C
106	PIP483.6	PASS	EC-6.2.9.1	**0.128**	13	2.97	T
81	PIP483.6	PASS	EC-6.2.9.1	**0.122**	9	0.04	T
107	PIP483.6	PASS	EC-6.2.9.1	**0.120**	8	3.03	C
55	PIP483.6	PASS	EC-6.2.9.1	**0.119**	13	0.01	C
109	PIP483.6	PASS	EC-6.3.3-662	**0.117**	9	3.84	C
34	PIP483.6	PASS	EC-6.3.3-662	**0.112**	12	1.19	C
77	PIP483.6	PASS	EC-6.2.9.1	**0.110**	8	0.02	C
70	PIP483.6	PASS	EC-6.2.9.1	**0.091**	10	0.30	T
66	PIP483.6	PASS	EC-6.2.9.1	**0.090**	10	0.42	T
87	PIP483.6	PASS	EC-6.2.9.1	**0.081**	10	0.01	T
91	PIP483.6	PASS	EC-6.2.9.1	**0.080**	10	0.08	C
120	PIP483.6	PASS	EC-6.2.9.1	**0.080**	12	0.38	T
84	PIP483.6	PASS	EC-6.3.3-662	**0.054**	8	2.80	C
113	PIP483.6	PASS	EC-6.3.3-662	**0.044**	8	1.53	C
114	PIP483.6	PASS	EC-6.3.3-662	**0.044**	12	1.53	C

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **4.113 kN** (Member 96, C). **Right-angle double coupler of Class A (10 kN slipping resistance)** shall be used for all scaffold steel tube connections at node joints.

Check: 4.113 kN < 10.0 kN → **PASS**

Member force table: see Appendix D.

Table C.1 — Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force $F_{s,k}$ (kN)	**10.0**	15.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.8
	Pull-apart force $F_{p,k}$ (kN)	20.0	30.0
	Rotational moment $M_{T,k}$ (kNm)	—	0.13

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	—	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	—	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.6

FRICTIONAL RESISTANCE

Max vertical reaction per support (Max f_y) = Resistance × 0.15 / C_F	= **5.96 kN**
Coefficient of friction (C_F)	= **0.3**
Frictional Resistance (FR) = C_F × Max f_y	= **1.788 kN**
Spring Stiffness = FR / 0.15	= **11.92 kN** ← applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results. Diagrams and displacement tables are in Appendix C and Appendix D.

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement — LC 17 Member 28 | Member length L = 1800.0 mm	**2.198 mm**	L/100 = 1800/100 = **18.0 mm**	**PASS**

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement — LC 19 Member 38 | Member length L = 1700.0 mm	**2.208 mm**	L/200 = 1700/200 = **8.5 mm**	**PASS**

CONCLUSION

The structural analysis and design of the **A Frame** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.39** at Member 121). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 18.0mm** and **L/200 = 8.5mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

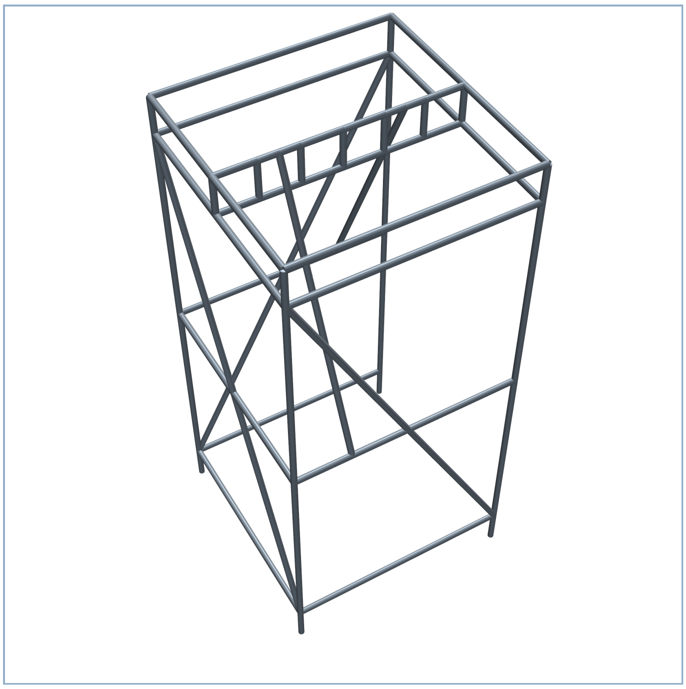
INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.

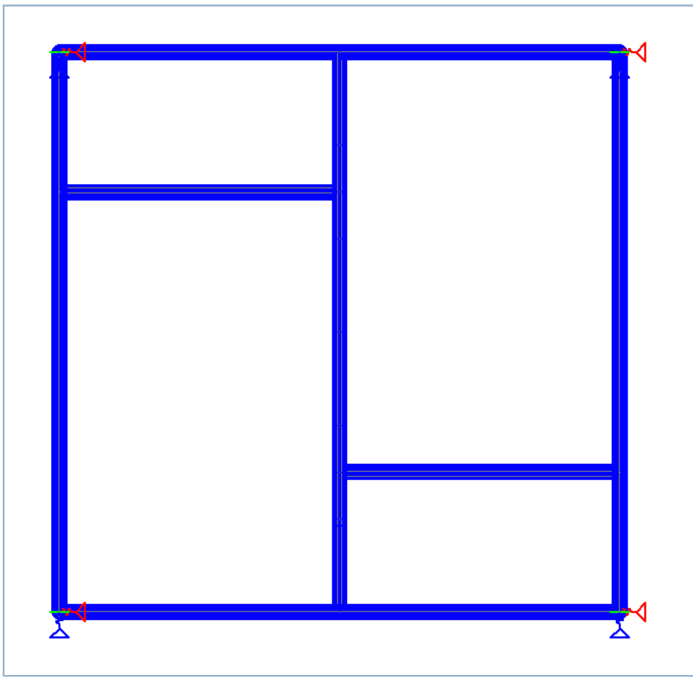
2. The work vicinity must be barricaded and appropriate signage placed at all access points.
3. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
4. A valid work permit must be obtained, discussed, and signed before erection commences.
5. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
6. Lay out and level the four base plates at the standard positions, ensuring a plan footprint of 1.8m x 1.8m in accordance with the approved working drawing.
7. Erect the four scaffold tube standards (48.3 x 3.6 mm) vertically on the base plates and secure with base collars. Check plumb in both axes before proceeding.
8. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to tie the four standards together and form the rigid base frame.
9. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
10. Install mid-level horizontal ledgers and transoms at 3.7m height in accordance with the working drawing.
11. Fix all vertical bracing members diagonally across the face panels of the A-frame structure using right-angle couplers to provide sway resistance in both X and Z axes as shown on the working drawing.
12. Install the 1800mm ladder beam horizontally across the top of the standards at 4.0m height, connecting to both standards with right-angle double couplers. Ensure the beam is level before fully tightening all couplers.
13. Rig the chain hoist to the ladder beam at the designated hoist point shown on the working drawing. Confirm the hoist is rated for a minimum safe working load equal to or greater than the design SWL.
14. The chain hoist attachment point and all coupler connections at ladder beam level shall be inspected and torque-checked by a competent person before any lifting operation commences.
15. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before any lift is performed.
16. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
17. All modifications made after erection must be carried out by a competent scaffold team with the written approval of the scaffold engineer. End-user or third-party modification of this scaffold structure is strictly prohibited.
18. For dismantling, perform the erection steps in reverse order.

APPENDIX A — MODEL VIEWS

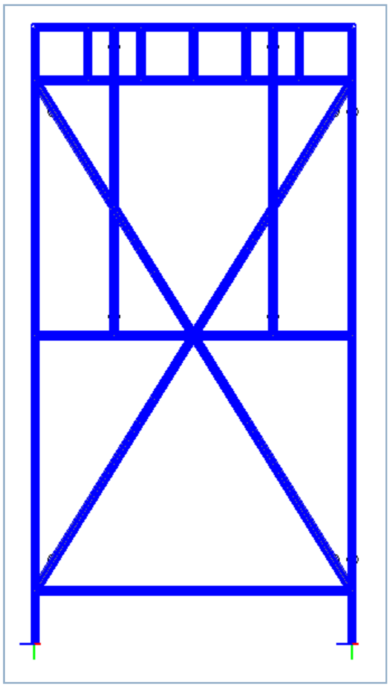
3D Isometric View



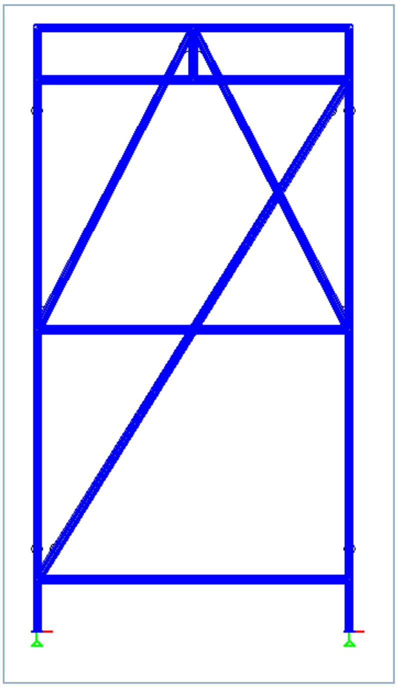
Plan View



Elevation from +X Axis



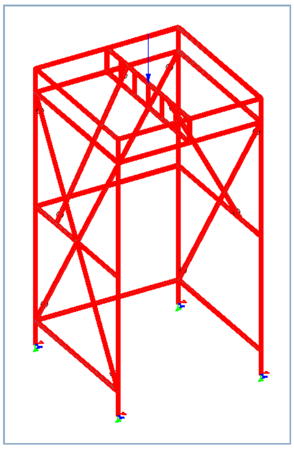
Elevation from +Z Axis



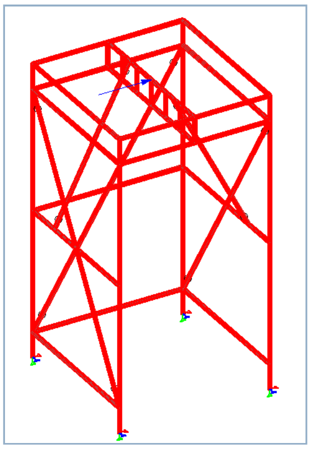
Scaffold Dimensions: W = 1.8m | D = 1.8m | H = 4.0m | **Lifts:** 0.3m, 2.0m, 3.7m, 4.0m

APPENDIX B — LOAD DIAGRAMS

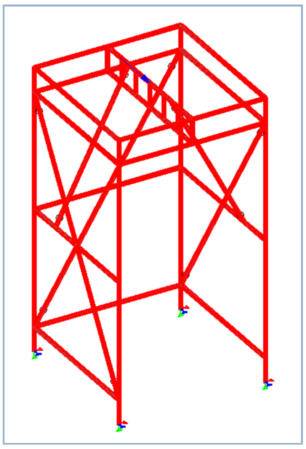
Chain Hoist Load (CL)



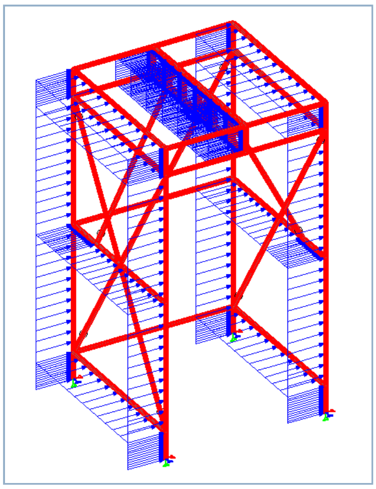
Impact Load — X Axis (ILX)



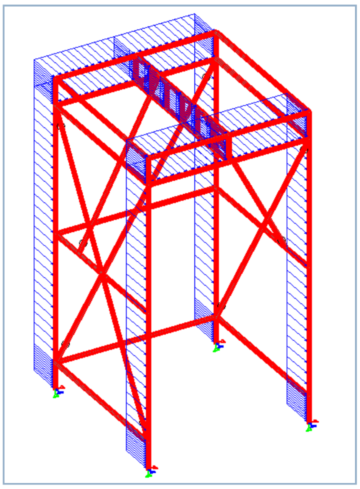
Impact Load — Z Axis (ILZ)



Wind Load — X Axis (WLX)

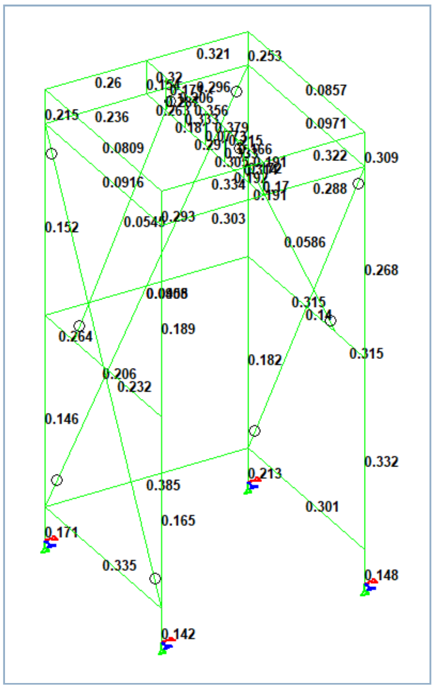


Wind Load — Z Axis (WLZ)

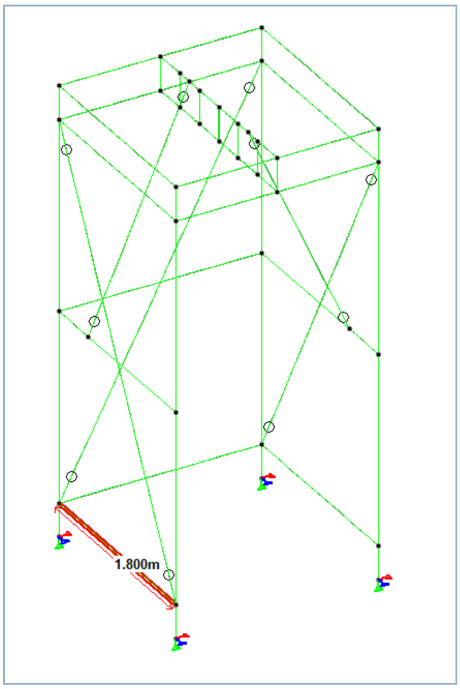


APPENDIX C — ANALYSIS RESULTS

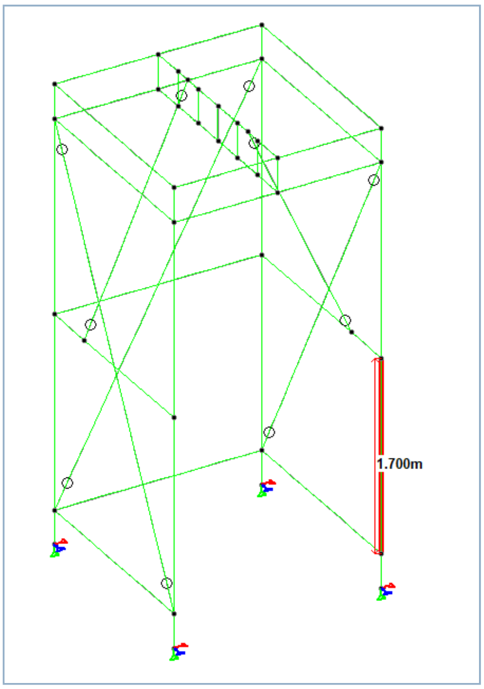
Utilization Ratio (UC) Diagram



Vertical Deflection Diagram



Horizontal Deflection Diagram



APPENDIX D — GUI RESULTS TABLES

Member Force / Code Check Table (Connection Stability)

Beam	L/C	Node	Axial Force kN	Shear-Y kN	Shear-Z kN	Torsion kN-m	Moment-Y kN-m	Moment-Z kN-m
96	16	45	5.824	-1.358	-0.122	-0.009	0.010	-0.277
96	20	45	5.108	-1.272	-0.123	-0.009	0.011	-0.264
96	14	45	5.092	-1.285	-0.123	-0.009	0.010	-0.266
103	16	36	5.084	-1.305	0.043	0.008	-0.010	0.262
75	20	48	5.080	1.284	0.046	0.009	0.013	-0.123
75	14	48	5.073	1.296	0.047	0.009	0.013	-0.125
104	14	51	5.015	1.376	-0.126	-0.009	0.039	0.133
75	16	48	5.011	1.368	0.046	0.009	0.013	-0.136
104	20	51	4.998	1.389	-0.126	-0.009	0.039	0.135
103	14	36	4.996	-1.379	0.043	0.008	-0.010	0.273
103	20	36	4.988	-1.390	0.043	0.008	-0.011	0.275
104	15	51	4.824	1.259	0.593	0.011	0.066	0.116
103	15	36	4.802	-1.252	-0.209	-0.011	-0.138	0.256
96	19	45	4.742	-1.274	-0.061	-0.013	0.024	-0.263
75	19	48	4.717	1.304	-0.030	0.013	-0.023	-0.129
96	15	45	4.686	-1.385	-0.692	-0.035	0.256	-0.279
75	15	48	4.664	1.440	0.095	0.035	-0.098	-0.153
104	19	51	4.662	1.367	-0.040	-0.012	0.026	0.134
103	19	36	4.637	-1.390	-0.059	0.012	-0.024	0.273
104	16	51	4.308	1.304	-0.125	-0.008	0.038	0.123
101	16	49	3.612	-1.515	-0.161	-0.003	-0.019	-0.204
109	16	54	3.612	-0.938	0.123	-0.003	-0.043	0.024
109	20	54	3.070	-0.854	0.119	-0.003	-0.043	0.023
101	20	49	3.070	-1.426	-0.162	-0.003	-0.019	-0.191
106	16	50	3.048	-1.103	-0.014	-0.000	-0.029	0.168
100	20	46	3.026	1.169	0.007	0.003	0.022	-0.169

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
28	2.150	1.050	17	837	42.660	-0.029	5.279
24	2.101	0.750	17	856	43.251	-0.037	-5.278
28	2.092	1.050	19	860	41.672	0.126	5.062
28	2.031	1.050	15	886	32.954	0.047	5.047
24	1.920	0.750	19	937	43.770	0.076	-5.072
24	1.772	0.750	15	1015	35.468	0.134	-5.027
120	1.097	0.900	15	1641	36.249	-0.143	1.125
121	1.069	0.450	15	1684	35.337	-0.328	3.543
120	1.054	0.900	19	1707	45.226	-0.135	1.044
121	1.026	0.450	19	1754	43.639	-0.449	3.449
45	0.875	0.900	15	1542	43.792	-1.722	-4.886
121	0.860	0.450	17	2092	43.947	-0.543	3.302
120	0.847	0.900	16	2126	-1.420	-0.085	31.325
120	0.815	0.900	20	2209	-1.457	-0.090	29.289
120	0.806	0.900	14	2232	-1.435	-0.100	0.788
45	0.700	0.900	19	1927	52.529	-1.232	-4.945
49	0.652	0.900	19	2070	47.592	-1.534	4.931
49	0.634	0.900	20	2128	-2.948	-1.246	28.243
49	0.620	0.900	14	2177	-2.921	-1.246	-0.254
49	0.614	0.900	16	2199	-2.953	-1.213	30.246
45	0.495	0.787	17	2726	51.161	-0.454	-5.147
49	0.431	0.675	17	3135	51.165	-0.385	5.162
121	0.425	1.050	16	4236	-0.315	-0.067	30.460
49	0.417	0.900	15	3236	38.188	-0.954	4.951
121	0.409	1.050	20	4395	-0.313	-0.074	27.338
121	0.407	1.050	14	4419	-0.318	-0.076	0.432
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Max = **2.198 mm** (LC 17) | L = Member 28 length (1800 mm) | Allowable L/100 = 18.0 mm → **PASS**

Horizontal Displacement Table (SLS Combinations)

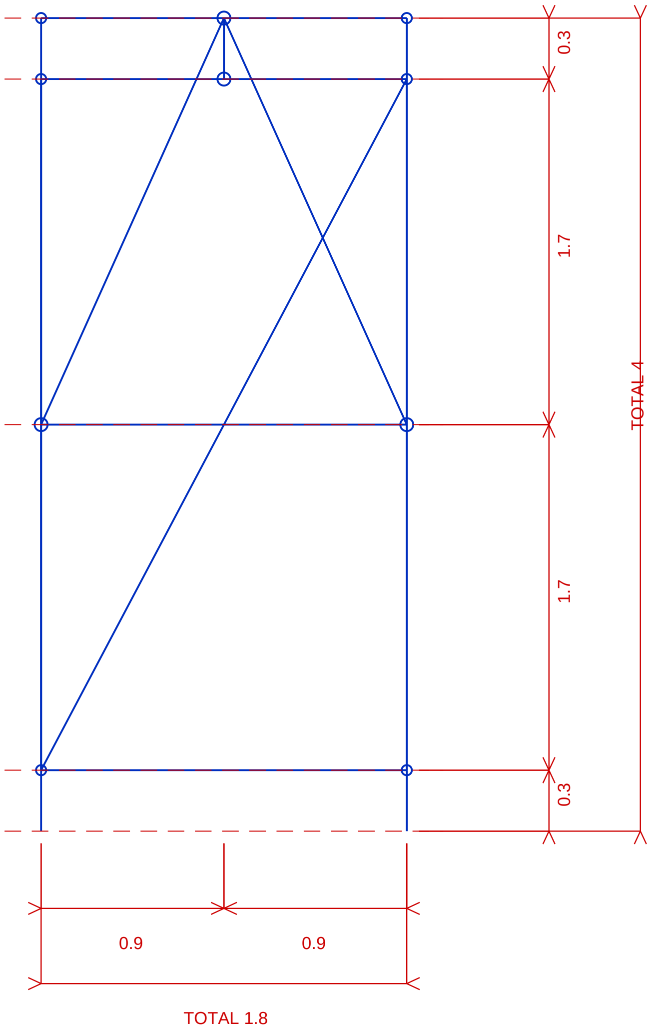
Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
38	2.349	0.992	19	723	52.142	-0.014	-4.557
38	2.067	0.992	15	822	43.380	-0.018	-4.297
38	1.835	0.992	17	926	49.375	0.004	-5.058
40	1.521	0.992	17	1117	47.285	-0.008	5.307
61	1.300	0.992	15	1307	46.013	-0.037	4.908
61	1.290	0.850	17	1317	56.607	-0.012	5.181
40	1.259	0.992	19	1350	44.603	-0.023	5.307
61	1.253	0.850	19	1356	54.530	-0.044	4.759
59	1.250	0.567	15	1359	49.567	-0.029	-5.519
59	1.249	0.567	19	1361	58.227	-0.023	-5.393
59	1.200	0.708	17	1416	57.257	0.012	-5.363
38	0.861	0.992	20	1974	2.901	-0.037	28.749
34	0.769	0.992	19	2210	45.086	-0.008	4.418
38	0.754	0.992	16	2255	3.054	-0.047	31.059
40	0.699	0.992	20	2432	-2.871	-0.023	28.217
36	0.696	0.850	17	2441	45.966	-0.038	-5.301
34	0.682	0.850	17	2491	45.952	0.011	5.096
36	0.677	0.850	19	2510	45.055	-0.052	-5.267
40	0.644	0.992	15	2640	34.153	-0.019	5.192
55	0.607	0.567	20	2798	-1.217	-0.024	28.327
38	0.595	0.992	14	2857	2.906	-0.025	0.547
40	0.562	0.992	16	3023	-2.986	-0.027	30.512
36	0.550	0.708	15	3089	36.170	-0.051	-5.274
55	0.505	0.567	19	3364	44.973	-0.018	5.493
34	0.479	1.133	14	3551	-1.211	-0.030	-0.700
34	0.469	0.850	15	3625	36.147	-0.002	4.690
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Max = **2.208 mm** (LC 19) | L = Member 38 length (1700 mm) | Allowable L/200 = 8.5 mm → **PASS**

APPENDIX E — REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment — Scaffolds — Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures — General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures — Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Dredger P.	How to Approximate 10-Minute Mean Wind Speed from 3-Second Gust	—
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 — Structural Analysis and Design Software	2022

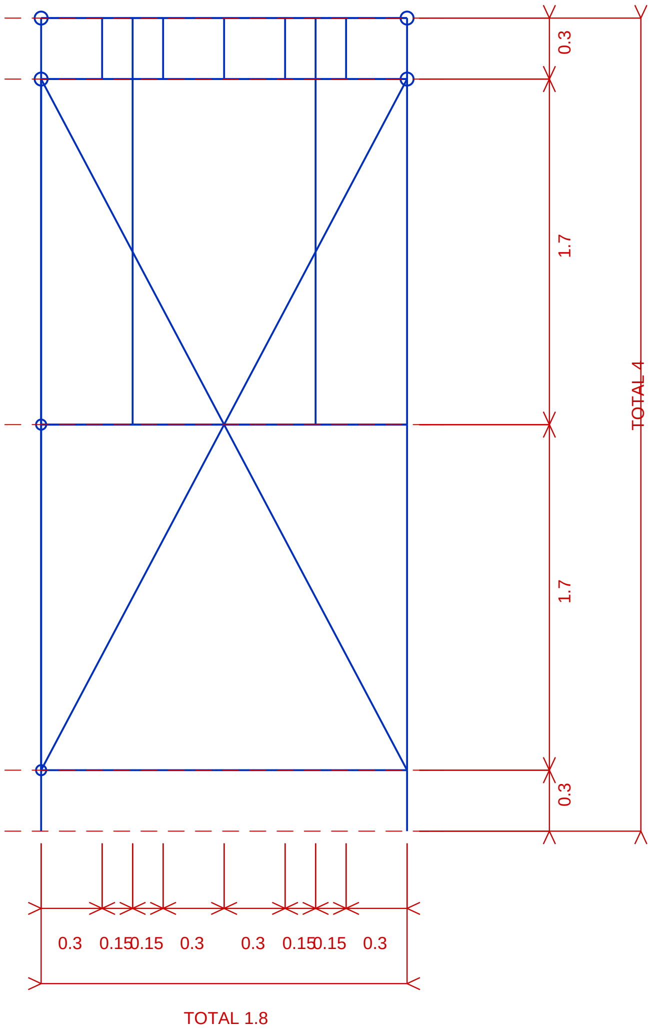
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.
Report: A-Frame Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



FRONT VIEW

PROJECT **SCAFFOLDING WORKS**		DRAWING NO. **PMS-WA-ARCO-003-FRONT-VIEW**		
DRAWING TITLE **SCAFFOLD GENERAL ARRANGEMENT - FRONT VIEW**		SCALE **NTS**	DATE **09-JUN-2026**	REV **0**
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED

COMPANY	
SHEET **1 OF 4**	SIZE **A4**



SIDE VIEW

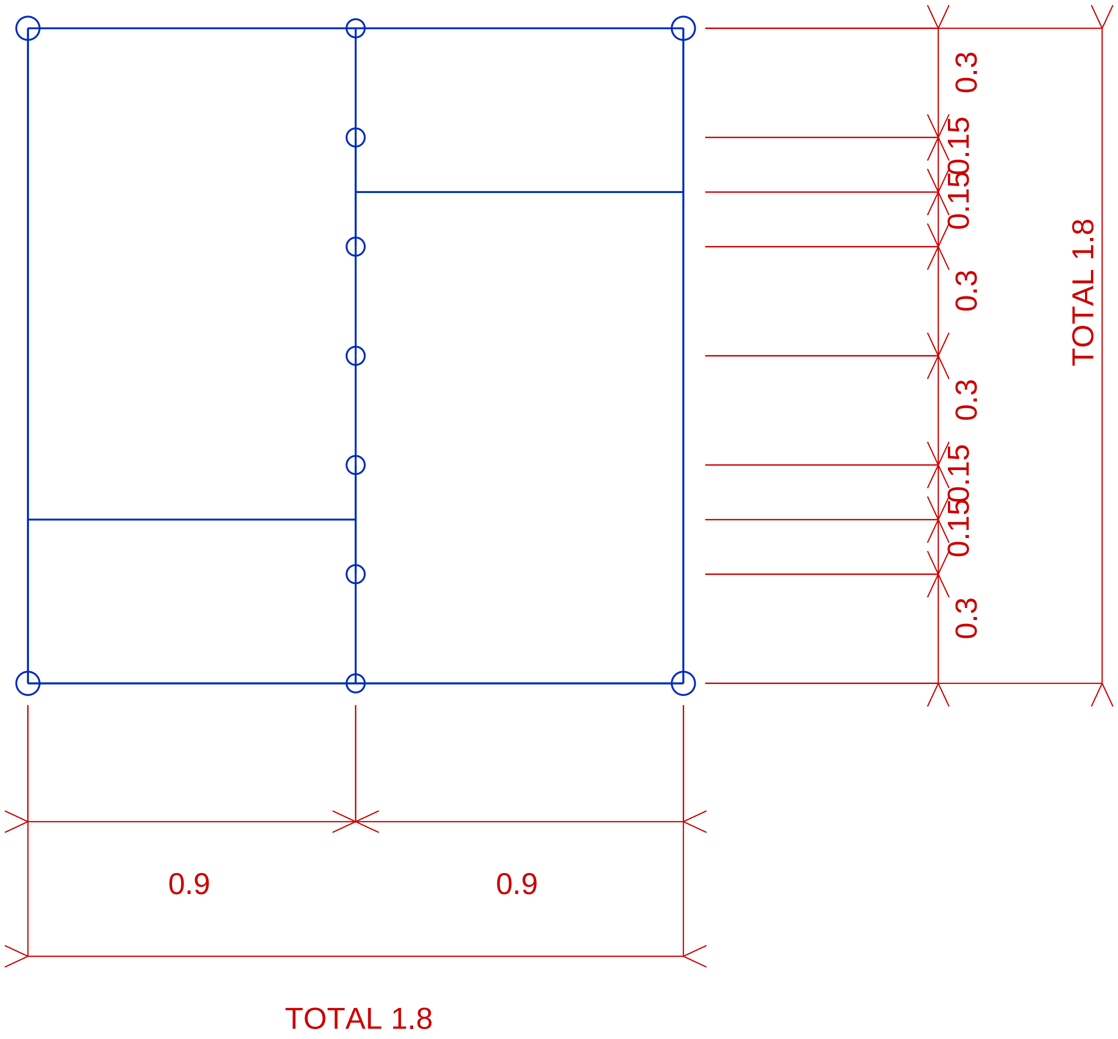
PROJECT **SCAFFOLDING WORKS**		DRAWING NO. **PMS-WA-ARCO-003-SIDE-VIEW**	
DRAWING TITLE **SCAFFOLD GENERAL ARRANGEMENT - SIDE VIEW**		SCALE **NTS**	DATE **09-JUN-2026**	REV **0**
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED

NLNG

ARCO M&E

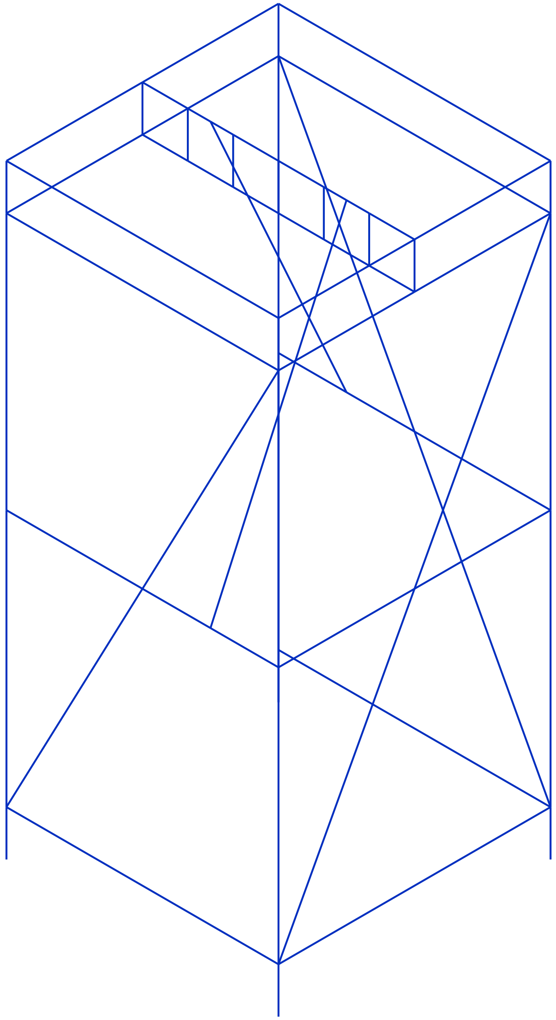
COMPANY

SHEET	SIZE
2 OF 4	A4



PLAN VIEW

PROJECT	SCAFFOLDING WORKS
DRAWING NO.	PMS-WA-ARCO-003-PLAN-VIEW
DRAWING TITLE	SCAFFOLD GENERAL ARRANGEMENT - PLAN VIEW
SCALE	NTS
DATE	09-JUN-2026
REV	0
CLIENT	
LOCATION	
DRAWN	Nnamani, Ifeanyi
CHECKED	
APPROVED	
COMPANY	NLNG / ARCO M&E
SHEET	3 OF 4
SIZE	A4



3D VIEW

PROJECT

SCAFFOLDING WORKS

DRAWING NO.

PMS-WA-ARCO-003-3D-VIEW

DRAWING TITLE

SCAFFOLD GENERAL ARRANGEMENT - 3D VIEW

SCALE	DATE	REV
NTS	**09-JUN-2026**	**0**

CLIENT	LOCATION

DRAWN	CHECKED	APPROVED
Nnamani, Ifeanyi		

COMPANY

SHEET	SIZE
4 OF 4	A4

3D Isometric View